

Branch: Mechanical Engg. (Second Year B.Tech.)

Subject with Subject Code:- Theory of Machines-I (BTMEC402)

Date:-28/11/2019

Sem:- IV

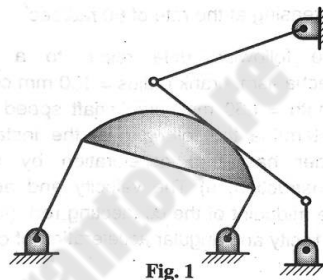
Marks: 60

Time:- 3 Hrs.

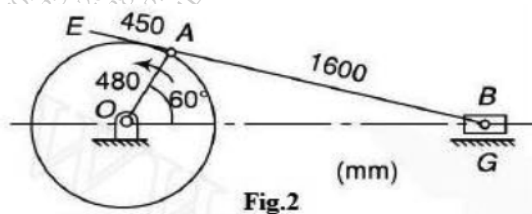
Instructions:-

1. Figures to the right indicate full marks.
2. Clearly mention the main question number along with the sub questions.
3. Assume suitable data, if necessary.
4. All questions are compulsory.

Q.No.	Question	Marks
1	Solve any Two :	
	a) i) Describe Watt's straight line generating mechanism	2.5
	ii) Find degrees of freedom for the mechanism as shown in the Fig.1.	2.5



- Explain the types of constrained motions with neat sketches. **05**
 - Explain with the help of neat sketches any two inversion of double slider crank chain. **05**
- 2 a) For the configuration of slider crank mechanism shown in Fig.2, Calculate **06**
- acceleration of the slider at B
 - acceleration of the point E
 - angular acceleration of the AB
- OA rotates at 20 rad/sec counter clockwise.



- In a horizontal IC engine mechanism, the crank of length 5 cm rotates at a uniform speed of 240 rpm. The length of connecting rod is 20 cm. when the crank has turned by 30° from its inner dead centre, locate all the instantaneous centers of the mechanism & find the angular velocity of the connecting rod & also the velocity of piston **04**

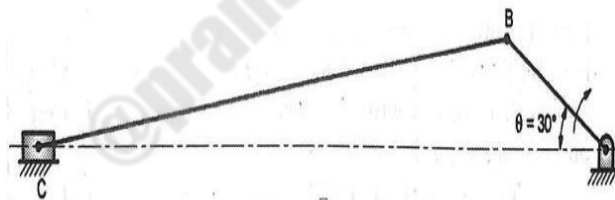


Fig.3.

3 Solve **Any Two**:

- A load of 15 kN is raised by means of screw jack. The mean diameter of the square threaded screw is 42 mm & the pitch is 10 mm. A force of 120 N is applied at the end of a lever to raise the load. Determine the length of lever to be used & mechanical advantage obtained. Is the screw self-locking? Take $\mu = 0.12$. 05
- What are uniform pressure & uniform wear theories? Deduce expressions for the friction torque considering both the theories for a flat collar. 05
- Deduce an expression for the efficiency of an inclined plane when a body moves down a plane. 05

4 Solve **Any Two**:

- An automotive single plate clutch consists of two pairs of contacting surfaces. The outer diameter of the friction disk is 270 mm. The coefficient of friction is 0.3 and the maximum intensity of pressure is 0.3 N/mm^2 . The clutch is transmitting a torque of 531 N-m. Assuming uniform wear theory, calculate: 05
 - the inner diameter of the friction disk; and
 - spring force required to keep the clutch engaged.
- A brake drum of 440 mm in diameter is used in a braking system as shown in Fig.4. The brake lever is inclined at an angle of 20° with the horizontal. A vertical force of 400 N magnitude is applied at the lever end. The coefficient of friction is 0.35. The brake drum has a mass of 160 kg & it rotates at 1500 rpm. Determine the 05
 - Braking torque
 - Number of revolution made by the drum & the time taken before coming to rest from the instant of brake is applied

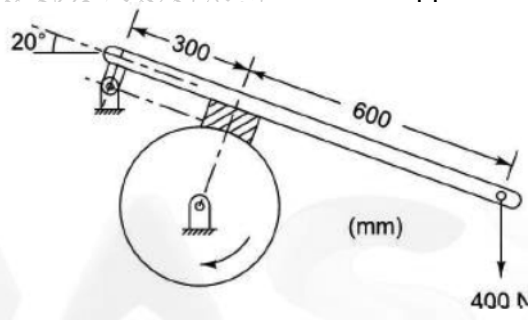


Fig.4.

- Discuss Prony brake dynamometer & Rope brake dynamometer with neat sketches. 05
- Draw the profile of a cam operating a knife-edge follower having lift of 30 mm. The cam raises the follower with uniform acceleration & deceleration for 120° of the rotation followed by a period of dwell for 30° . The follower descends for the next 90° rotation of the cam with SHM, again followed by 08

a dwell period. The cam rotates at a uniform speed of 800 rpm & has a least radius of 30 mm. What will be the maximum velocity & acceleration of the follower during the lift & the return?

b) Write various types of cams.

02

6 Solve **Any Two**:

a) Explain the direct & reverse crank method for determining unbalanced forces in radial engines.

05

b) A three cylinder radial engine running at 1500 r.p.m. is having its axes at 120° to each other. The stroke is 120 mm & each connecting rod is 215 mm long. The mass of reciprocating parts is 3 kg per cylinder. Determine the primary & secondary unbalanced force acting on the engine.

05

c) Explain the method of balancing of several masses rotating in same planes.

05

*****Paper End*****

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular End Semester Examination – Summer 2022

Course: B. Tech. Branch: Mechanical Engineering Semester: IV

Subject Code & Name: BTMC402 Theory of Machines I

Max Marks: 60

Date: 18/08/22

Duration: 3.45 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q.1 Solve the following.

- A) Fig. 1 shows part of an opposed piston engine mechanism. The velocity of the piston E for the given instant is 780 mm/s. the crank OA rotates at a uniform speed in a clockwise direction and makes an angle of 45° to vertical as shown in Fig. 1. Draw a velocity diagram and determine the speed of the crank in rpm. OA= 50mm, AB= 200 mm, BC= 110 mm, CD= 220 mm and DE= 100 mm. Apply 6

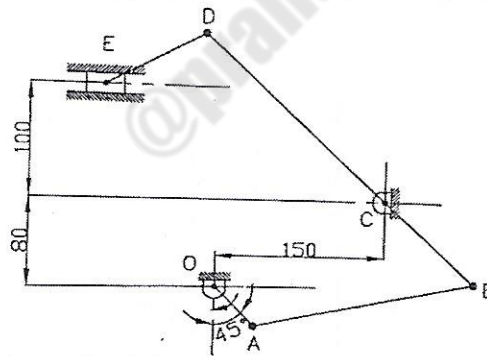


Fig. 1

- B) Explain with sketches different types of constrained motions. Remember 6

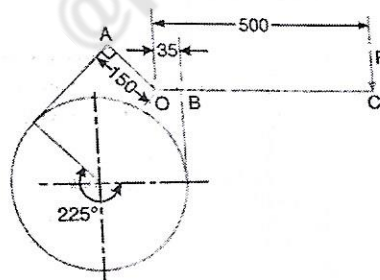
Q.2 Solve Any Two of the following.

- A) A shaft has a number of collars integral with it. The external diameter of the collar is 400 mm and the shaft diameter is 250 mm. if the intensity of the pressure is 0.35 N/mm^2 and the coefficient of friction is 0.05. Estimate power absorbed when the shaft runs at 105 rpm carrying a load of 150 kN and a number of collars required. Apply 6
- B) Derive the expression for a total torque acting on a truncated conical pivot bearing considering uniform wear, Remember 6
- C) A conical pivot supports a load of 20 kN, the cone angle is 120° and the intensity of normal pressure is not to exceed 0.3 N/mm^2 . The external Apply 6

diameter is twice the internal diameter. Find the outer and inner radii of the bearing surface. If the shaft rotates at 200 rpm and the coefficient of friction is 0.1. Find the power absorbed in friction. Assume uniform pressure.

Q. 3 Solve Any Two of the following.

- A) A centrifugal clutch is to transmit 15 kW at 900 rpm. The shoes are four in number. The speed at which the engagement is $3/4^{\text{th}}$ of the running speed. The inside radius of the pulley rim is 150 mm and the center of gravity of the shoe lies at 120 mm from the center of the spider. The shoes are lined with Ferrodo for which the coefficient of the friction may be taken as 0.25. Determine the mass of the shoes and size of the shoes, if the angle subtended by the shoes as the center of the spider is 60° and the pressure exerted on the shoes is 0.1 N/mm^2 . Apply 6
- B) A differential band brake as shown in Fig. 2 has an angle of contact of 225° . Apply 6
The band has a compressed woven lining and bears against a cast iron drum of 350 mm diameter. The brake is to sustain a torque of 350 Nm and the coefficient of friction between the band and drum is 0.3. Find the necessary force 'P' for the clockwise rotation of the drum and the value of 'OA' for the brake to be self-locking when the drum rotates in clockwise.



All dimensions in mm.

Fig. 2

- C) Differentiate between absorption type dynamometers and transmission type dynamometers. Remember 6

Q.4 Solve the following.

- A) Draw the profile of a cam operating a roller reciprocating follower having a lift of 40 mm. the roller diameter is 20 mm. the minimum radius of the cam is 30 mm. the cam raises the follower with simple harmonic motion for 110° of its rotation followed by a period of dwell for 80° . the follower descends for the next 120° rotation of the cam follower with uniform acceleration and deceleration followed by a dwell period. Apply 6

B) Explain with figures any three types of followers.

Remember

6

Q. 5 Solve Any Two of the following.

A) Explain the direct and reverse crank method for determining unbalanced forces in radial engines.

Remember

6

B) Explain the method of balancing of several masses in the same planes.

Remember

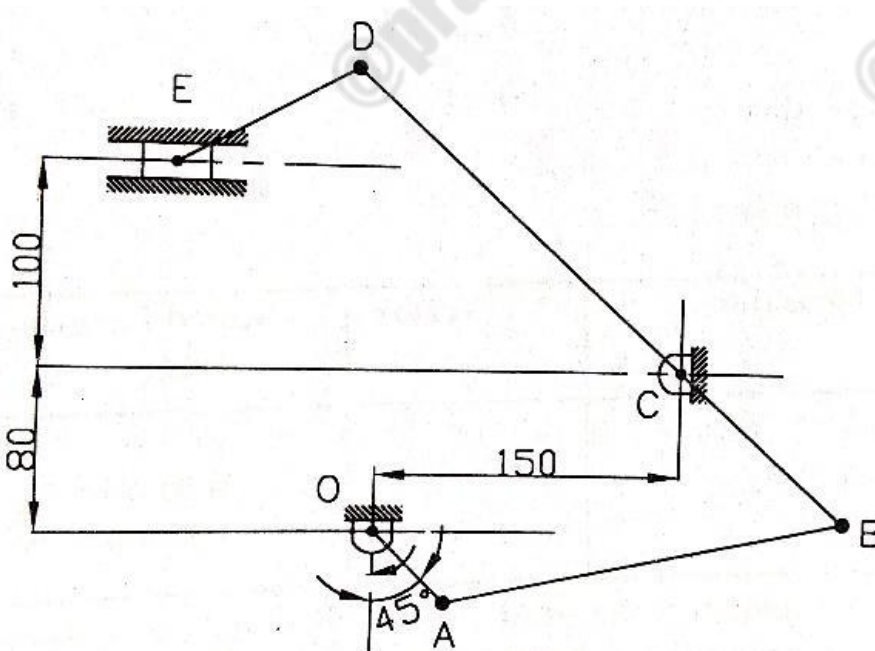
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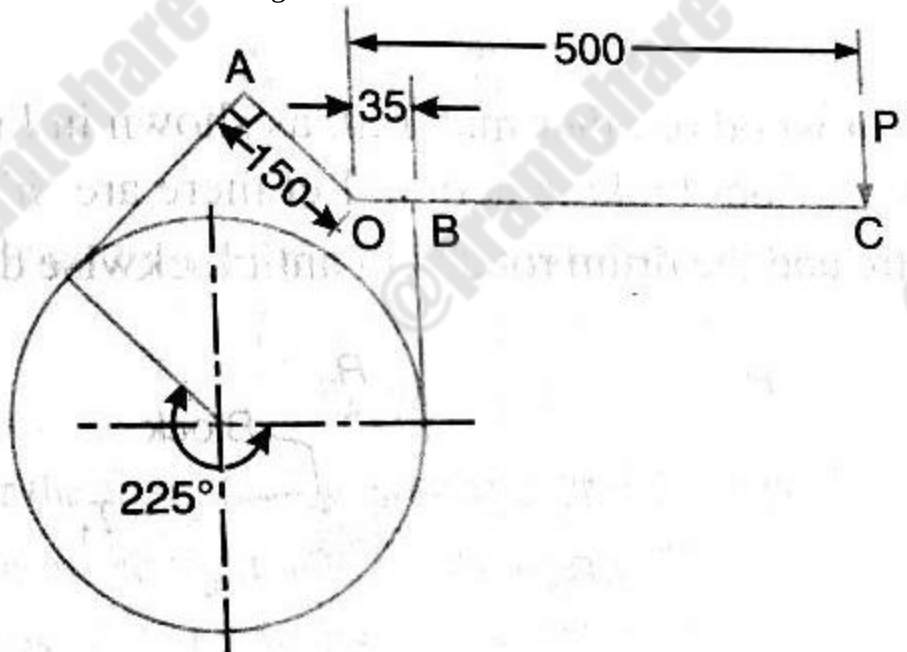
C) A four-cylinder vertical engine has cranks 150 mm long. The planes of rotation of the first, second and fourth cranks are 400 mm and 200 mm respectively from the third crank and their reciprocating masses are 50 kg, 60 kg and 50 kg respectively. Find the mass of the reciprocating parts for the third cylinder and the relative angular positions of the cranks in order that the engine may be in complete primary balance.

Apply

6

***** End *****

	DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Examination – Summer 2022 Course: B. Tech. Branch :Mech.Engg Semester :IV Subject Code & Name: BTMEC402 Theory of Machines I Max Marks: 60 Date: Duration: 3 Hr.	
	Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.	
		(Level/CO) Marks
Q. 1	Solve Any Two of the following. (This is just a sample instruction)	
A)	Fig. 1 shows part of an opposed piston engine mechanism. The velocity of the piston E for the given instant is 780 mm/s. the crank OA rotates at a uniform speed in a clockwise direction and makes an angle of 45° to vertical as shown in Fig. 1. Draw a velocity diagram and determine the speed of the crank in rpm. OA= 50mm, AB= 200 mm, BC= 110 mm, CD= 220 mm and DE= 100 mm.  Fig No 01	6
B)	Explain with sketches different types of constrained motions.	6
C)	How Kinematic pairs are classified ? Explain with neat Diagram	6

Q.2	Solve Any Two of the following. (This is just a sample instruction)		
A)	Derive the expression for a total torque acting on a truncated conical pivot bearing considering uniform wear,		6
B)	A shaft has a number of collars integral with it. The external diameter of the collar is 400 mm and the shaft diameter is 250 mm. if the intensity of the pressure is 0.35 N/mm ² and the coefficient of friction is 0.05. Estimate power absorbed when the shaft runs at 105 rpm carrying a load of 150 kN and a number of collars required.		6
C)	Write a short note on Friction types with suitable example.		6
Q. 3	Solve Any One of the following. (This is just a sample instruction)		
A)	A differential band brake as shown in Fig. 2 has an angle of contact of 225°. The band has a compressed woven lining and bears against a cast iron drum of 350 mm diameter. The brake is to sustain a torque of 350 Nm and the coefficient of friction between the band and drum is 0.3. Find the necessary force 'P' for the clockwise rotation of the drum and the value of 'OA' for the brake to be self-locking when the drum rotates in clockwise.  All dimensions in mm. Fig No 02		6
B)	Differentiate between absorption type dynamometers and transmission type dynamometers		6
Q.4	Solve Any Two of the following. (This is just a sample instruction)		
A)	How cams are classified? Explain in details with neat sketches.		6
B)			6

	Draw the profile of a cam operating a roller reciprocating follower having a lift of 40 mm. the roller diameter is 20 mm. the minimum radius of the cam is 30 mm. the cam raises the follower with simple harmonic motion for 110° of its rotation followed by a period of dwell for 80°. the follower descends for the next 120° rotation of the cam follower with uniform acceleration and deceleration followed by a dwell period.		
C)	Explain with figures any three types of followers.		6
Q. 5	Solve Any One of the following. <i>(This is just a sample instruction)</i>		
A)	A four-cylinder vertical engine has cranks 150 mm long. The planes of rotation of the first, second and fourth cranks are 400 mm and 200 mm respectively from the third crank and their reciprocating masses are 50 kg, 60 kg and 50 kg respectively. Find the mass of the reciprocating parts for the third cylinder and the relative angular positions of the cranks in order that the engine may be in complete primary balance.		6
B)	Explain the method of balancing of several masses in the same planes		6
	*** End ***		

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Dr. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE
Regular / Supplementary End Semester Examination – Summer 2023

Course: **B. Tech.** Branch: **Mechanical Engineering** Semester: **IV**

Subject Code & Name: **BTMC402 Theory of Machines - I**

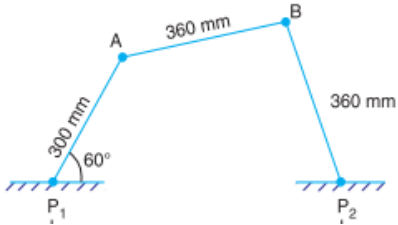
Max Marks: **60**

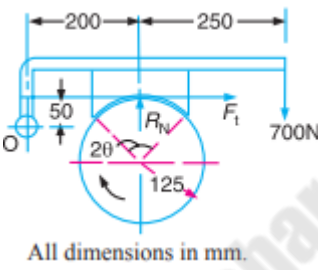
Date: **15/07/23**

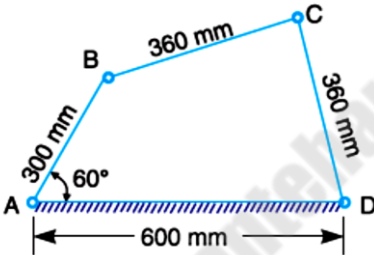
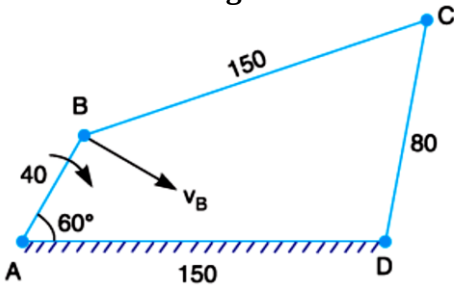
Duration: **3.45 Hr.**

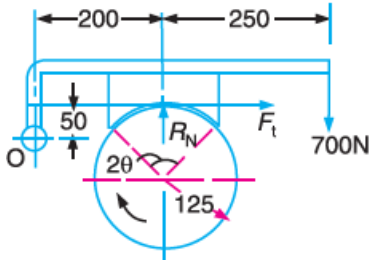
Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Solve Any TWO of the following.		
A)	Explain in brief the inversions of Four Bar Chain.	Remember	06
B)	An engine mechanism is shown in figure 1. The crank CB = 100 mm and the connecting rod BA = 300 mm with centre of gravity G, 100 mm from B. In the position shown, the crankshaft has a speed of 75 rad/s and an angular acceleration of 1200 rad/s². Find: 1. Velocity of G and angular velocity of AB, and 2. Acceleration of G and angular acceleration of AB.  Fig. 1	Apply	06
C)	The dimensions and configuration of the four bar mechanism, shown in Fig. 2, are as follows: P₁A = 300 mm; P₂B = 360 mm; AB = 360 mm, and P₁P₂ = 600 mm. The angle AP₁P₂ = 60°. The crank P₁A has an angular velocity of 10 rad/s and an angular acceleration of 30 rad/s², both clockwise. Determine the angular velocities and angular accelerations of P₂B, and AB and the velocity and acceleration of the joint B.  Fig. 2	Apply	06
Q.2	Solve Any TWO of the following.		
A)	State the laws of (i) Static friction ; (ii) Dynamic friction	Understanding	06

B)	conical pivot supports a load of 20 kN, the cone angle is 120° and the intensity of normal pressure is not to exceed 0.3 N/mm^2 . The external diameter is twice the internal diameter. Find the outer and inner radii of the bearing surface. If the shaft rotates at 200 r.p.m. and the coefficient of friction is 0.1, find the power absorbed in friction. Assume uniform pressure.	Apply	06
C)	Derive the expression for a total torque acting on a conical pivot bearing considering uniform wear.	Analyze	06
Q.3	Solve Any <u>TWO</u> of the following.		
A)	With neat sketch, explain the Multi plate clutch.	Understanding	06
B)	A single block brake is shown in Fig. 3. The diameter of the drum is 250 mm and the angle of contact is 90°. If the operating force of 700 N is applied at the end of a lever and the coefficient of friction between the drum and the lining is 0.35, determine the torque that may be transmitted by the block brake.  All dimensions in mm. Fig 3	Apply	06
C)	Discuss the Prony Brake Dynamometer and Rope brake Dynamometer with neat sketches.	Create	06
Q.4	Solve the following.		
A)	Explain the different types of followers.	Understanding	06
B)	Construct the profile of a cam to suit the following specifications : Least radius of cam = 50 mm ; Diameter of roller = 25 mm; Angle of lift = 90° ; Angle of Dwell = 45°; Angle of fall = 75° ; Lift of the follower = 24.5 mm ; Number of pauses are two of equal interval between motions. During the lift, the motion is S.H.M. During the fall the motion is uniform acceleration and deceleration. The speed of the cam shaft is uniform.	Analyze	06
Q.5	Solve Any <u>TWO</u> of the following.		
A)	Explain the method of balancing of several masses in the same planes.	Understanding	06
B)	Explain the direct and reverse crank method for determining unbalanced forces in radial engines.	Understanding	06
C)	Four masses m_1 , m_2 , m_3 and m_4 are 200 kg, 300 kg, 240 kg and 260 kg respectively. The corresponding radii of rotation are 0.2 m, 0.15 m, 0.25 m and 0.3 m respectively and the angles between successive masses are 45° , 75° and 135° . Find the position and magnitude of the balance mass required, if its radius of rotation is 0.2 m.	Apply	06

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Winter Examination – 2023 Course: B.Tech. Branch : Mech. Engg./Mech. Engg. Sandwich Sem :IV Subject Code & Name: BTMC402 - Theory of Machines-I Max Marks: 60 Date: 18 /01/2024 Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve any 2 from the following questions.		12
A)	Discuss double slider crank chain mechanism & its inversion	CO1	6
B)	Fig. shows $AB = 300$ mm, $BC = CD = 360$ mm, and $AD = 600$ mm. The angle $BAD = 60^\circ$. The crank AB rotates uniformly at 100 r.p.m. Locate all the instantaneous centres and find the angular velocity of the link BC 	CO1	6
C)	In a four bar chain $ABCD$, AD is fixed and is 150 mm long. The crank AB is 40 mm long and rotates at 120 r.p.m. clockwise, while the link $CD = 80$ mm oscillates about D. BC and AD are of equal length. Find the angular velocity of link CD when angle $BAD = 60^\circ$. 	CO1	6
Q.2	Solve any 2 from the following questions.		12
A)	Describe uniform pressure theory for flat pivot bearing	CO2	6
B)	A conical pivot supports a load of 20 kN, the cone angle is 120° and the intensity of normal pressure is not to exceed 0.3 N/mm². The external diameter is twice the internal diameter. Find the outer and inner radii of the bearing surface. If the shaft rotates at 200 r.p.m. and the coefficient of friction is 0.1, find the power absorbed in friction. Assume uniform pressure.	CO2	6
C)	Explain in brief Hydrostatic and hydrodynamic lubrication.	CO2	6
Q. 3	Solve any 2 from the following questions.		12
A)	Explain in brief different types of absorption type dynamometers	CO3	6

B)	A single block brake is shown in Fig. The diameter of the drum is 250 mm and the angle of contact is 90°. If the operating force of 700 N is applied at the end of a lever and the coefficient of friction between the drum and the lining is 0.35, determine the torque that may be transmitted by the block brake. 	CO3	6
C)	Explain the construction & working of centrifugal Clutch	CO3	6
Q.4	Solve any 2 from the following questions.		12
A)	Explain with figures any three types of followers.	CO4	6
B)	A cam is to give the following motion to a knife-edged follower : 1. Outstroke during 60° of cam rotation; 2. Dwell for the next 30° of cam rotation; 3. Return stroke during next 60° of cam rotation, and 4. Dwell for the remaining 210° of cam rotation. The stroke of the follower is 40 mm and the minimum radius of the cam is 50 mm. The follower moves with uniform velocity during both the outstroke and return strokes. Draw the profile of the cam when (a) the axis of the follower passes through the axis of the cam shaft, and (b) the axis of the follower is offset by 20 mm from the axis of the cam shaft.	CO4	6
C)	Explain any three terms used in cam along with sketch.	CO4	6
Q. 5	Solve any 2 from the following questions.		12
A)	Explain the method of balancing of several masses in the same planes	CO5	6
B)	A shaft carries four masses A, B, C and D of magnitude 200 kg, 300 kg, 400 kg and 200 kg respectively and revolving at radii 80 mm, 70 mm, 60 mm and 80 mm in planes measured from A at 300 mm, 400 mm and 700 mm. The angles between the cranks measured anticlockwise are A to B 45°, B to C 70° and C to D 120°. The balancing masses are to be placed in planes X and Y. The distance between the planes A and X is 100 mm, between X and Y is 400 mm and between Y and D is 200 mm. If the balancing masses revolve at a radius of 100 mm, find their magnitudes and angular positions.	CO5	6
C)	Explain the direct and reverse crank method for determining unbalanced forces in radial engines.	CO5	6
	*** End ***		

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Regular and Supplementary Summer 2024 Course: B. Tech. Branch : Mechanical Engineering Semester : IV Subject Code & Name: BTMC402 Theory of Machines - I Max Marks: 60 Date: 14/06/2024 Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	Write types of ICR and explain Kennedy's theorem with example	1	6
B)	The radius of crank and length of connecting rod are 100 mm and 400 mm respectively. If the crank rotates clockwise with the angular velocity of 10 rad/s and is at 45 degree from inner dead center. Find velocity of slider using relative velocity Method.	2	6
C)	In a slider crank mechanism the radius of crank and length of connecting rod are 150 mm and 600 mm respectively. If the crank rotates clockwise with the angular velocity of 12 rad/s and is at 45 degree from inner dead center. Find velocity of slider and angular velocity of connecting rod using Klien's construction method.	2	6
Q.2	Solve Any Two of the following.		12
A)	Write concept of friction axis	1	6
B)	In a thrust bearing the external and internal diameters of the contacting surfaces are 416 mm and 286 mm respectively. The total axial load is 125 KN and intensity of pressure 0.7 N/mm^2 . The shaft rotates at 370 RPM. Taking coefficient of friction 0.04 calculate power lost in overcoming the friction.	2	6
C)	Derive an expression for frictional torque of flat collar bearing assuming uniform wear theory	2	6
Q. 3	Solve Any Two of the following.		12
A)	A multi plate clutch has 6 active friction surfaces. Power transmitted is 20KW at 400 RPM. Inner and outer radii of friction surface are 90mm and	2	6

	120mm respectively. Assuming uniform wear and coefficient of friction 0.3. Find the maximum intensity of pressure between discs.		
B)	Derive an expression for torque transmitted by cone clutch due to friction considering uniform pressure theory	2	6
C)	Determine the axial force required to engage cone clutch transmitting 25 KW power at 250 RPM. Average friction diameter of cone is 400 mm, semi cone angle 10° and coefficient of friction 0.25	2	6
Q.4	Solve Any Two of the following.		12
A)	Derive an expression for braking torque for single block brake when the line of action of tangential braking force is at distance 'a' below the fulcrum 'O' at clockwise rotation of drum.	2	6
B)	In a rope brake dynamometer Diameter of flywheel 1m, Diameter of rope 10mm, Dead weight of the brake 50Kg, Speed of Engine 180 rpm, spring balance reading 120 N, Find brake power of the Engine	3	6
C)	A torsion dynamometer is fitted on turbine shaft to measure the angle of twists. It is observed that the angle of twist. It observed that the shaft twists in 1.5° in a length of 5m at 500 rpm. The shaft is solid and has diameter of 200 mm. Modulus of rigidity of shaft material is 85 GPa. Find the power transmitted by turbine.	3	6
Q. 5	Solve Any Two of the following.		12
A)	Explain with neat sketch types of cam	1	6
B)	A cam rotating at uniform speed required to give the Knife edge follower motions as: (i) Follower to move outwards through the distance of 30mm during 90° of cam rotation with uniform velocity (ii) Follower to dwell for next 60° of cam rotation (iii) Follower to return to its original position during 90° cam rotation with uniform velocity (iv) Follower to dwell for remaining period. The minimum radius of the cam is 50 mm. Draw the cam profile when follower is offset by 20mm from the axis of the cam shaft	3	6
C)	Four masses m_1 , m_2 , m_3 and m_4 are 200 kg, 300 kg, 240 kg and 260 kg	3	6

	respectively. The corresponding radii of rotation are 0.2 m, 0.15 m, 0.25 m and 0.3 m respectively and the angles between successive masses are 45° , 75° and 135° . Find the position and magnitude of the balance mass required, if its radius of rotation is 0.2 m		
	*** End ***		